

PROGRAMMING IN JAVA

CSA0920

Assignment 1

Question: HashMap vs Hashtable

Write a Java program that creates both a HashMap and a Hashtable, and attempts to insert a null key or a null value into each. Display the outcome of each attempt (success or exception/error) to demonstrate that HashMap permits null keys/values while Hashtable does not. Concepts: HashMap, Hashtable, null handling.

Code:

```
import java.util.HashMap;
import java.util.Hashtable;

public class Main {
    public static void main(String[] args) {

        HashMap<Integer, String> hm = new
        HashMap<>();

        hm.put(null, "Java");
        hm.put(1, null);
```

```

        System.out.println("HashMap : " + hm);

        Hashtable<Integer, String> ht = new
        Hashtable<>();

        try {
            ht.put(null, "Java");
        } catch (Exception e) {
            System.out.println("Hashtable null key: " + e);
        }

        try {
            ht.put(1, null);
        } catch (Exception e) {
            System.out.println("Hashtable null value: " + e);
        }
    }
}

```

Output:

```

HashMap : {null=Java, 1=null}
Hashtable null key: java.lang.NullPointerException:
Cannot invoke "Object.hashCode()" because "key" is null
Hashtable null value: java.lang.NullPointerException

```